SMART ENVIRONMENTAL DASHBOARD

TI SCHOOL • Smart Environmental Intelligence System

Project overview: A school-focused environmental monitoring dashboard hosted by an ESP32 and viewed through a browser over the local Wi-Fi network. It is designed to make environmental conditions easier to observe and discuss.

Purpose

To help staff observe classroom comfort and environmental conditions, notice potential issues, and consider practical actions such as improving ventilation or reducing avoidable noise.

Sensors and measurements

Sensor / module	What it measures or contributes
SCD30 CO■ sensor	Carbon dioxide (CO■), temperature and relative humidity
TSL2591 light sensor	Ambient light level, reported in lux
GY-SGP40 VOC sensor	VOC-related signal for indoor air-quality awareness
DY038 sound sensor	Sound-level signal for noise monitoring; calibrated dB(A) requires suitable calibration
DHT22 temperature/humidity sensor	Temperature and relative humidity (where included in the build)
BMP180 pressure sensor	Atmospheric pressure (where included in the build)

System structure

Sensors Environmental inputs	**ESP32** Reads/serves configured data	**Wi-Fi browser dashboard** Status, trends and recommendations

Dashboard features and use

- **At-a-glance status:** visual condition labels and indicators.
- **Environmental overview:** sensor metrics supported by the connected hardware and active firmware.
- **Trends and recommendations:** supports observation and practical classroom actions.
- Power the ESP32, connect it to the configured Wi-Fi, then open its dashboard address from a browser on the same network.

Measurement note: A sensor contributes data only when it is physically connected and enabled in the running firmware. If demonstration values are being used, they are illustrative—not live measurements. Sound-sensor output must not be described as calibrated dB(A) unless calibrated. This prototype is an environmental awareness aid, not a certified safety instrument.